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Evaluating energy supply service reliability for commercial air conditioning loads from the distribution network aspect

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HIGHLIGHTS

- Defined the "Loss of Load" for air-conditioning loads considering customer comfort.
- Developed systematic reliability indices and method to quantify load reliability.
- An improved two-stage control method for air-conditioning loads demand response.
- Proposed method considers both demand response and load rebound suppression

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ABSTRACT

The limited amount of dispatching flexibility has encouraged the development and implementation of demand response in the distribution system. Commercial air conditioning loads are promising demand response resources and tend to have greater potential for centralized regulation. From the perspective of distribution networks, evaluating the reliability of energy supply services for air conditioning loads should be of much concern because reliable services would directly affect the customers' tendency to demand response. However, traditional load reliability evaluation does not fully explore the fact that the energy supply processes of air conditioning loads can be flexible. Modeling the energy demands as flexible loads, a reliability assessment algorithm for commercial air conditioning loads is proposed for distribution systems in this paper. The "loss of load" event is innovatively defined considering customer preferences. Novel systematic indices are established that are specialized for describing the reliability of energy supply services. Afterward, a two-stage control strategy is proposed considering the comfort requirements of customers, which would affect reliability performance. The first stage optimizes the operating state of the air conditioning system internal equipment collaboratively, while the second stage sacrifices some comfort for sufficient utilization of the thermal self-storage capacity and load rebound suppression. Finally, the proposed method is validated in a standard test system with voltage violation risks. The results demonstrate the impacts of different control strategies on the reliability performance of commercial air conditioning loads. Analysis drawn from the sensitivity of load reliability indices can assist distribution system operators in coordinating energy supply reliability and dispatching requirements.

1. Introduction

The economic development of a country is tightly coupled with the energy demand from its customers, especially the demand for electricity. Demand response (DR) is an advanced load-side resource management technology that allows customers to participate in the dispatching-operation control via changing the electric usage when the system reliability is at risk [1,2]. The reliability of power systems has become one of the focus issues raised by DR technologies [3–5]. The reason is that many traditional loads are augmented with loads that can interact and respond to grid conditions, and this approach might introduce both new opportunities and challenges into power systems due to the volatility and uncertainty of electricity demand.

In particular, among the various load-side resources, air conditioning loads account for almost 40–60% of a building's total electricity consumption [6]. The demand for such thermostatically controlled appliances can be shifted due to their thermal self-storage ability. Considerable efforts have already been devoted to the

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Nomenclature

Sets and Indices

i	footnote representing the ith zone of a CAC system
m	footnote representing the mth CAC system in a district
k	footnote representing the system failure mode
t	index of a time snapshot
Δt	the control interval
Ω_s	the set of commercial CAC systems
Ψ_m	the set of zones of the mth commercial CAC system
ch, pump fan, coil	footnote representing the parameters of the chiller, fan, water pump and coil unit

Variables

δ	the acceptable percentage of discomfort time for customers
ξ_t	the membership degree to describe customer comfort at different temperatures
χ_{expect}	the expected comfort membership degree
$\lambda_t^{\text{loadrate}}$	load rate of the central air conditioning system at time tth
τ_i^{change}	the change time of the ith terminal unit
$t_{\text{DR,start}}$	time when a DR event occurs
$t_{\text{DR,end}}$	time when a DR event ends
m_t^{water}	the water flowrate of a chiller
m_t^{air}	air volume of a supply fan
$m_t^{\text{air-new}}$	air volume of a CAC system
$T_{\text{o},t}^{\text{ch}}$	outlet chilled water temperature
$T_{\text{o},t}^{\text{coil}}$	outlet air temperature
$T_{\max}$	the upper bound of the room temperature
$T_{\min}$	the lower bound of the room temperature
T_i^{room}	the room temperature of the ith zone

Parameters

n	total number of zones in a CAC system
c_{water}	the specific heat capacity of water
c_{air}	the specific heat capacity of air
$\text{COP}_{\text{rated}}$	rated coefficient of performance
Q_{rated}, P_{rated}	rated cooling capacity and electric power
$m_{\text{rated}}^{\text{air}}$	rated maximum air volume of a fan
$m_{\text{rated}}^{\text{water}}$	rated chilled water flowrate
E_g	the heat exchange efficiency
$Q_i^{\text{H/C}}$	the heating/cooling power of zone i
$P_t^{\text{cooling-cycle}}$	total power of a cooling water circulation

Functions

$\Gamma(X, y)$	selecting the smallest y elements in the set X
$\lceil\cdot\rceil$	rounding up
$\psi(\cdot)$	probability of a stochastic event

applications of DR strategies to improve system security and reliability in the last decade [7–9]. As an example, the potential of air conditioning loads following the generation of a considerable scale of renewable energy has been proven in [10]. The same problem was also investigated in [11] using a coordination algorithm for air conditioning loads and batteries to smooth the tie-line power fluctuations caused by renewable energy. Ref. [12] studied the reliability of a high renewable penetration microgrid considering a coordination strategy of air conditioning systems. Ref. [13] developed a more advanced short-term reliability evaluation algorithm with DR provided by air conditioning loads. The DR strategies for air conditioning loads have also proven to be effective for improving power quality such as peak shaving [14], voltage regulation [15], and frequency control [16].

To achieve these system reliability enhancements, the reliable energy supply in the DR duration is an essential factor that depends mainly on the willingness of customers [17]. One of the obstacles is that customers have high requirements for energy supply reliability, resulting in low-risk tolerance for new technologies. From the distribution network aspect, load reliability evaluation is also an essential task for distribution system operators (DSOs) in the power system planning and operation. Intrinsically, reliability is a measure of the ability to supply power to load users according to acceptable standards, including continuity and quantity in power systems. For a flexible and controllable load, the acceptable standard is obviously different from a traditional load, and an unclear definition might lead DSOs to over- or under-estimate the responsiveness of flexible loads. Thus, power supply reliability for some flexible loads such as EVs, has attracted much attention, and corresponding evaluation indices are used in the process of continuous improvement [18,19]. Ref. [20] takes into account the customer satisfaction constraints for non-concrete controllable loads while evaluating the reliability of active distribution networks. However, there is little literature specifically studying the power supply reliability for air conditioning loads themselves in the DR duration, while only some reliability indices for traditional loads are considered in the system reliability assessment [13].

In addition, the adjustable capacity of air conditioning loads and the customer experience of energy supply services are both determined by DR strategies. Coordinated control of the decentralized air conditioning loads of residential or public buildings has received much attention [11,15,21,22]. Most of these strategies are based on the instantaneous startup and shutdown of the system in direct load control. However, this control mode is not suitable for commercial central air conditioning (CAC) systems in which the chiller units are screw-type or centrifugal. Their life span will be reduced by frequent periodic start-stop control [23]. The power consumption of CAC systems is related not only to the compressor but also to multiple heat exchange modules such as the chilled water circulation system, cooling water circulation system, and fresh air system [24].

Recent studies indicate there are many optimization problems and control methods for commercial CAC systems [25,26]. Some efforts have been made in recent years to demonstrate CAC systems' ability to respond to grid conditions. For example, the characteristics of the fan [27], pump [28] and chiller [29] in a commercial CAC system have been utilized to participate in frequency regulation. Ref. [24] gives a hierarchical coordinated control method to handle all the internal equipment mentioned above. All of the previous works neglect the complex thermodynamics of each zone in a building. In fact, the thermal power consumption of building zones can also be adjusted to incorporate the overall optimization of the air conditioning system. Regarding this issue, duty cycling control for decentralized air conditioning loads is introduced into the terminal equipment of each floor in the CAC systems for load reduction in [23]. Ref. [30] also proposes a hierarchical control scheme to consider the temperature control of terminal units. Ref. [31] utilizes transactive control to optimize a commercial building air conditioning system for DR considering the zone thermal model. However, once the thermal demand is adjusted, it will inevitably cause the deviation between the actual temperature and customer's expected temperature. It should be a last resort to adjust the comfort level of customers, and this priority is often neglected in the above literature.

Motivated by this background, a novel reliability evaluation algorithm is proposed for commercial air conditioning loads from the viewpoint of the distribution system. New reliability indices for energy supply services are established because the reliability indices for conventional loads are not applicable for flexible air conditioning loads. Subsequently, a two-stage control strategy considering the comfort requirements of customers is also proposed in this paper, since the control strategy for DR is considered a necessary input to the reliability evaluation. The proposed method and control strategy are deployed to alleviate the voltage violation risk of the distribution network. These case studies indicate that better control effects can be achieved by considering the thermal storage capacity of a commercial CAC system as well as the impacts of different DR control strategies on the load reliability. The main contributions of this paper are as follows.

(1) The reliability of commercial air conditioning loads is defined and evaluated. Considering the acceptable standard of flexible loads, the *Loss of Load* event is defined innovatively for air conditioning loads, in which different preferences can be considered. Customer-oriented and energy-oriented indices are developed to quantify load reliability.
(2) An improved two-stage DR strategy for commercial air conditioning loads are proposed comprising global optimization of the equipment and adjustment of the thermal power consumption. Customers' comfort requirements with the energy supply services can be fully guaranteed.
(3) The first stage of the proposed method is the collaborative optimization of equipment operation state without sacrificing customers' comfort. The second stage is thermal demand optimization via an improved algorithm considering the strategies for the response and load rebound.

This paper is organized as follows. Section 2 presents the models of a typical commercial CAC system. Section 3 introduces the main reliability evaluation algorithm. Section 4 discusses the specific control strategy for demand response. Case studies are provided in Section 5. Finally, conclusions and future work are summarized in Section 6.

2. Commercial CAC system modeling

It is critical to combine the zone thermal demand to model the electricity consumption of a commercial CAC system. This section introduces the electric power model and the thermal behavior model of a commercial CAC system.

2.1. Electric power model of a commercial CAC system

The configuration of a typical commercial CAC system is shown in Fig. 1 and is mainly composed of some chiller units, chilled water pumps, cooling towers, cooling water pumps, ventilation units, fan coil units and so on. An electric power model of each equipment needs to be established to calculate the system power consumption. Detailed descriptions of these main components can be found in [32,33].

2.1.1. Chiller model

The chiller unit is the cold source of a commercial CAC system and is the major energy-consuming facility. According to the EnergyPlus manual, we model the chiller power P_t^{ch} using several polynomial functions as follows:

$$P_t^{\text{ch}} = Q_{\text{rated}}^{\text{ch}} * f_1(T_{o,t}^{\text{ch}}) * f_2(T_{o,t}^{\text{ch}}, \lambda_t^{loadrate})/\text{COP}_{\text{rated}}, \tag{1}$$

where f_1 is the function relation between the cooling capacity and the extracting water temperature of the chiller $T_{o,t}^{\text{ch}}$, and f_2 is utilized to determine the energy efficiency ratio (EER) using the extracting water temperature $T_{o,t}^{\text{ch}}$ and load rate $\lambda_t^{loadrate}$ of the chiller as follows:

$$\begin{cases} f_1 = a_1 + a_2 * T_{o,t}^{\text{ch}} + a_2 * (T_{o,t}^{\text{ch}})^2 + a_3, \\ f_2 = [b_1 + b_2 * T_{o,t}^{\text{ch}} + b_3 * (T_{o,t}^{\text{ch}})^2 + b_4] * [c_1 + c_2 * \lambda_t^{loadrate} + c_3 \\ \quad * (\lambda_t^{loadrate})^2], \end{cases} \tag{2}$$

In the above formulas, $a_{1,2,3}$, $b_{1,2,3,4}$ and $c_{1,2,3}$ are the chiller characteristic coefficients, and the load rate $\lambda_t^{loadrate}$ can be expressed using (3)

$$\lambda_t^{loadrate} = \frac{c_{\text{water}} m_t^{\text{water}} \, |T_{o,t}^{\text{ch}} - T_{\text{in},t}^{\text{ch}}|}{Q_{\text{rated}}^{\text{ch}} * f_1(T_{o,t}^{\text{ch}})}, \tag{3}$$

where $T_{\text{in},t}^{\text{ch}}$ is the water inlet temperature.

2.1.2. Supply fan and water pump model

The supply fan is a device that powers air circulation in an air conditioning system. Its power consumption P_t^{fan} is determined by the supply air rate m_t^{air} as follows [33]:

$$P_t^{\text{fan}} = \frac{m_{\text{rated}}^{\text{air}} * f_3(m_t^{\text{air}}) * \Delta P}{1000 \eta_{\text{fan}} * \rho_{\text{air}}}, \tag{4}$$

$$f_3 = d_1 + d_2 * \left(\frac{m_t^{\text{air}}}{m_{\text{rated}}^{\text{air}}}\right) + d_3 * \left(\frac{m_t^{\text{air}}}{m_{\text{rated}}^{\text{air}}}\right)^2 + d_4 * \left(\frac{m_t^{\text{air}}}{m_{\text{rated}}^{\text{air}}}\right)^3, \tag{5}$$

where f_3 represents the partial load factor of the fan, ΔP is the design pressure, η_{fan} is the efficiency, and $d_{1,2,3,4}$ are the fan characteristic coefficients.

The power of the water pump P_t^{pump} is relatively simple:

$$P_t^{\text{pump}} = e_1 * (m_t^{\text{water}} / m_{\text{rated}}^{\text{water}}) * P_{\text{rated}}^{\text{pump}}. \tag{6}$$

where e_1 is the water pump characteristic coefficient.

2.1.3. Coil unit model

Coil units are the key device for connecting the air circulation and the chilled water circulation. Because of the heat exchange between the air side and the water side of coil units, the air circulation and the chilled water circulation are not independent of each other. This relationship can be described by the coil unit efficiency model,

$$E_g = \frac{T_{\text{in},t}^{\text{coil}} - T_{o,t}^{\text{coil}}}{T_{\text{in},t}^{\text{coil}} - T_{o,t}^{\text{ch}}}, \tag{7}$$

where $T_{\text{in},t}^{\text{coil}}$ and $T_{o,t}^{\text{coil}}$ are the inlet and outlet air temperatures of a coil unit, respectively.

In fact, E_g is determined by m_t^{water} and m_t^{air}. A more detailed description of its calculation method can be found in [32]. Here, we focus on the calculation of the inlet air temperature; $T_{\text{in},t}^{\text{coil}}$ can be expressed as follow,

$$T_{\text{in},t}^{\text{coil}} = \frac{T_t^{\text{return\_air}}(m_t^{\text{air}} - m_t^{\text{air\_new}}) + T_t^{\text{out}} m_t^{\text{air\_new}}}{m_t^{\text{air}}}, \tag{8}$$

where $T_t^{\text{return\_air}}$ is the return air temperature.

The temperature of each zone is $T_t^{i,\text{room}}$, while the thermal load and supply air flowrate are $Q_t^{i,\text{room}}$ and $m_t^{i,\text{air}}$, respectively. Assuming that the thermal load entering the ceiling is $\kappa^i Q_t^{i,\text{room}}$, the return air temperature can be calculated via Eq. (9).

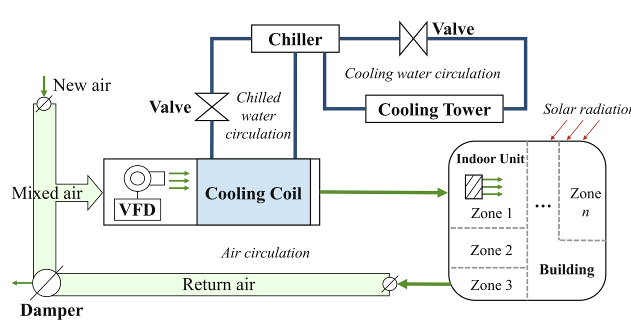

Fig. 1. Typical configuration of a commercial air conditioning system, including air circulation, chilled water circulation and cooling water circulation.

$$T_t^{\mathrm{return\_air}} = \frac{\sum_{i\in\Psi}(T_t^{i,\mathrm{room}*}m_t^{i,\mathrm{air}} + \kappa^i Q_t^i/c_{air})}{m_t^{\mathrm{air}}}. \tag{9}$$

2.2. Thermal behavior model of indoor zones

There are two accurate methods for modeling the thermal behavior of an indoor zone. One is known as the equivalent thermal parameters (ETP) model [34,35], while the other is to calculate the thermal load based on the law of energy conservation [23,36].

In this paper, we adopt the thermal burden coefficient method to establish a first-order differential equation and obtain the zone thermodynamic model as follows [23]:

$$X_i\cdot\frac{dT_i^{\mathrm{room}}(t)}{dt} + B_i\cdot T_i^{\mathrm{room}}(t) - (A_i - Q_i^{\mathrm{H/C}}) = 0, \tag{10}$$

$$T_i^{\mathrm{room}}(t_{n+1}) = \frac{A_i(t_n) - Q_i^{\mathrm{H/C}}(t_n)}{B_i} - \left(\frac{A_i(t_n) - Q_i^{\mathrm{H/C}}(t_n)}{B_i} - T_i^{\mathrm{room}}(t_n)\right)e^{-\frac{B_i}{X_i}\Delta t}. \tag{11}$$

where $T_i^{\mathrm{room}}(t_n)$ is the indoor temperature at time t_n, A_i is a time-dependent coefficient that is the sum of the hourly thermal burden, B_i is a constant coefficient in a certain building environment that is related to the heat transfer coefficient and area of the envelope structure, and X_i is constant, subject to the heat storage capacity of the zone's air or wall. The specific calculation methods for coefficients such as A_i, B_i and X_i are described in [23] and are not repeated here. A zone's heat transfer process is shown in Fig. 2.

3. Main algorithm and reliability indices

The *Loss of Load* event of air conditioning loads is defined and two reliability indices are proposed for the energy supply service of commercial CAC systems. This section shows how these indices are established.

3.1. General description of the process

As shown in Fig. 3, some dispersive commercial CAC systems can take part in the DR when some distribution system failures occur, such as the overvoltage or overloading problem caused by load transfer. To assess the energy supply service reliability of air conditioning loads, some necessary information is required as follows:

(1) *Control strategies of commercial CAC systems*: The adjustable potential of air conditioning loads is directly determined by the control strategies. The customer experiences vary for different control strategies, and this variation will have an essential effect on the reliability evaluation of energy supply services. The detailed control strategy is discussed in Section 4.
(2) *Following signals for demand response*: The load-following signals describe the DR requirements of the grid as determined by scenarios of distribution system failures. The adjustment of the air conditioning loads is jointly affected by both these signals and control strategies.
(3) *Distribution system failure information*: The failure information is usually statistically analyzed to form reliability indices such as the failure rate and duration of electrical equipment. These failures can cause a DR event of air conditioning loads and threaten the reliability of energy supply services.

3.2. Defining LoL for commercial air conditioning loads

In conventional reliability evaluations, LoL means that the continuity or quantity of power supply services to customers cannot be fully met. However, as a typical load that can be reduced, air conditioning loads can be adjusted by partially sacrificing some comfort. The requirements of the continuity and quantity are not necessary due to the thermal storage capacity. Ignoring this fact leads to an inaccurate estimation of load reliability. Therefore, the reliability of a power supply for such flexible loads needs to be redefined.

This paper defines the LoL in the energy supply services for air conditioning loads as follows. The quality of service that customers care most about is the guarantee of indoor temperature. Once the temperature setpoint $T_i^{\mathrm{setpoint}}(t)$ is fixed, maintaining the indoor temperature $T_i^{\mathrm{room}}(t)$ at this level is what a customer most needs since too low/high temperatures might bother this customer. A fuzzy membership degree ξ_t^i has been established to describe the customers' comfort at different temperatures, as shown in Fig. 4. Temperature preferences might vary among different customers and are reflected in the parameter choices of α_t, β_t and $T_i^{\mathrm{setpoint}}(t)$. If the membership degree ξ_t^i is less than the limit χ_{expect}^i, we assume that the temperature at time t_n is uncomfortable for this customer and mark this moment, as shown in Eq. (12):

$$f(\xi_t^i) = \begin{cases} 1, & \xi_t^i = 1\Big/\left[1 + \left|\left(\frac{T^{\mathrm{room}}(t) - T_i^{\mathrm{setpoint}}(t)}{\alpha_t}\right)^{2\beta_t}\right|\right] \leqslant \chi_{\mathrm{expect}}^i \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

Then, Eq. (13) presents the mathematical expression of the definition. It is used to examine whether LoL events happen in the DR duration.

$$\varepsilon_i = \begin{cases} 1, & \text{if}\left(\sum\limits_{t\in[t_{\mathrm{DR,start}},t_{\mathrm{DR,end}}]} f(\xi_t^i)\cdot\Delta t - \delta\cdot T_{\mathrm{respond}}\right) > 0 \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where $\varepsilon_i = 1$ means that LoL occurs for zone i and T_{respond} is the duration. The relationship between reliability and DR lies in the indoor temperature $T_i^{\mathrm{room}}(t)$. This relationship is not difficult to understand because the smaller the indoor temperature fluctuation to the setpoint, the better the reliability will be.

3.3. Establishing reliability indices for energy supply service

The commercial CAC system is designed to supply cooling service for many zones $i = 1, 2\ldots,n$. The first reliability index is a customer-oriented index named the annual thermal demand unsatisfaction indice (ATDUI, %). According to the definition of LoL, ATDUI can be calculated using

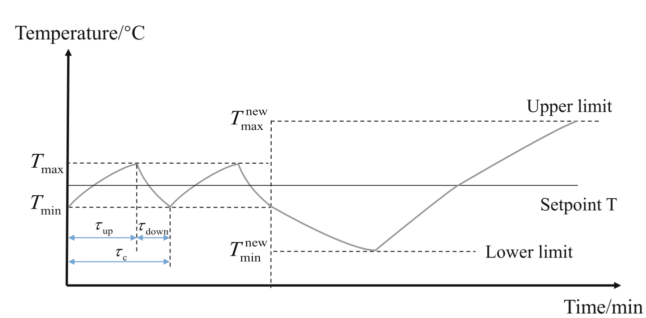

Fig. 2. Thermal behavior of an indoor unit. The room temperature would fluctuate between $[T_{min}, T_{max}]$ via thermostat controllers. Once the deadband of thermostat controllers is updated, a new temperature fluctuation range will be generated like $[T_{min}^{new}, T_{max}^{new}]$.

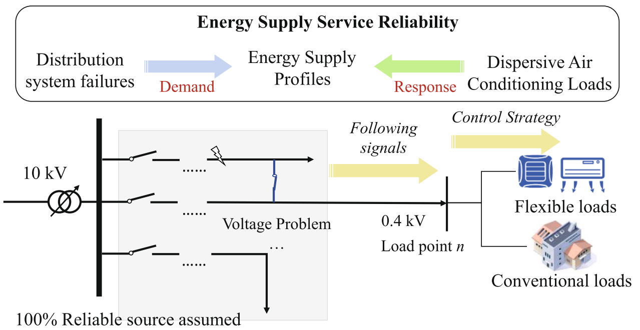

Fig. 3. Algorithm overview for evaluating the energy supply service reliability of air conditioning loads.

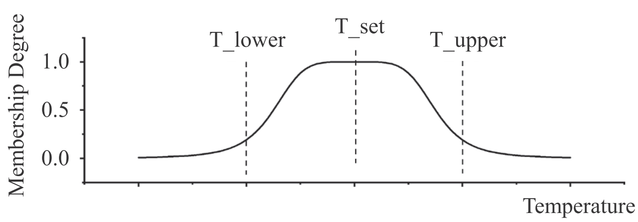

Fig. 4. The membership degree between temperature and comfort.

$$\mathrm{ATDUI} = \frac{\text{No. of LoL events}}{\text{No. of zones in the commercial CAC systems}} = \left[\sum_{k=1,2,\ldots}\sum_{m\in\Omega_s}\left(\psi_{k,m}\left(t_{\mathrm{DR,start}},\, t_{\mathrm{DR,end}}\right)\cdot\sum_{i\in\Psi_m}\varepsilon_i\right)\right]\Big/\sum_{m\in\Omega_s}N_m. \tag{14}$$

where $\psi_{k,m}(t_{\mathrm{DR,start}}, t_{\mathrm{DR,end}})$ denotes the probability that a DR event occurs at the mth commercial CAC system during $[t_{\mathrm{DR,start}}, t_{\mathrm{DR,end}}]$ by each failure mode k.

Another index is proposed oriented to the energy demand named the qualified energy not supplied indice (QENS, kWh/a). The difference between the ATDUI and QENS index is that the qualified electric power not supplied P_i^{NS} for zone i,

$$\mathrm{QENS} = \sum_{k=1,2,\ldots}\sum_{m\in\Omega_s}\left[\psi_{k,m}\left(t_{\mathrm{DR,start}},\, t_{\mathrm{DR,end}}\right)\sum_{i\in\Psi_m}\varepsilon_i P_i^{\mathrm{NS}}\Delta t\right], \tag{15}$$

$$P_i^{\mathrm{NS}} = \sum_{\xi_t^i\in\Gamma\left(\Phi_i,w=\left[\sum_{t\in[t_{\mathrm{DR,start}},t_{\mathrm{DR,end}}]} f(\xi_t^i)-\delta\cdot T_{\mathrm{respond}}/\Delta t\right]\right)} \frac{Q_i^{\mathrm{NS}}(\xi)}{\mathrm{COP}}. \tag{16}$$

where Φ_i is the set of ξ_t^i during $[t_{\mathrm{DR,start}}, t_{\mathrm{DR,end}}]$, and w indicates the number of the time interval with unqualified power supply to meet the required percentage of discomfort time δ.

In this paper, the cooling power Q^{H} of each zone is considered fixed during the cooling and warming periods. Therefore, $Q_i^{\mathrm{NS}}(\xi)$ is calculated by subtraction between $Q_{i,\mathrm{cooling}}^{\mathrm{H}}$ and $Q_{i,\mathrm{warming}}^{\mathrm{H}}$. In addition, we assume that the electrothermal conversion efficiency is constant when calculating P_i^{NS} using the thermal demand, i.e., $\mathrm{COP} = \mathrm{COP}_{\mathrm{rated}}$.

4. Control strategy of commercial air conditioning loads for demand response

The advent of two-way communication technology and advanced metering infrastructure (AMI) has further encouraged the implementation of demand response [37]. These advances in technology bring about flexibility and allow for the interaction of customers with utilities. Load aggregators can obtain the schedule from a superior dispatching center, and control operating states of commercial CAC systems to utilize their adjustable potential. This section introduces a

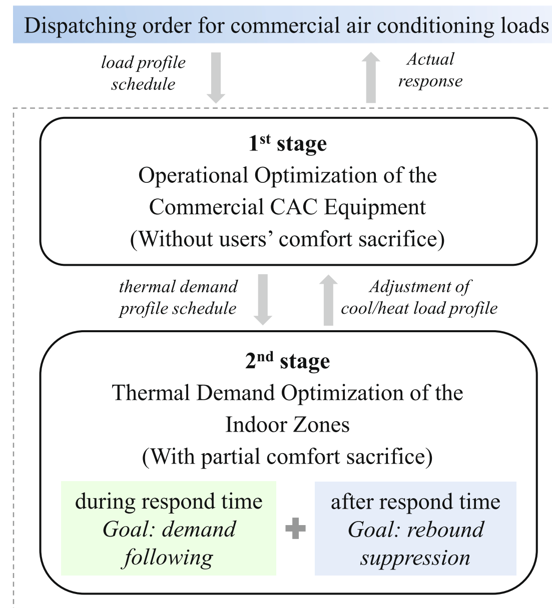

Fig. 5. The two-stage control strategy of a commercial CAC system for demand response.

two-stage control strategy to respond to grid demand for commercial air conditioning loads, as shown in Fig. 5.

(1) *The first stage*: the object is to meet scheduling requirements as much as possible via optimizing the operating states of the internal equipment collaboratively, such as the chilled water flow rate and air volume. The thermal demand of each zone is not adjusted in this step. Once the response requirement cannot be met by only optimizing the air conditioning system state, an adjustment schedule of thermal demand would be generated. The problem can be solved by building an optimization model.
(2) *The second stage*: some comfort sacrifice might be made for the control of terminal unit states. The change-time-priority-list method is used to ensure that the room temperature fluctuates within a specified deadband. Considering a disorderly setting of the temperature return behavior, this stage takes into account two strategies for demand response and rebound suppression. The object of the DR strategy is to follow the adjustment schedule generated from the previous stage, while the other is to control load rebound within an acceptable range.

4.1. First-stage scheduling method for the commercial AC equipment

Due to the complexity of the commercial CAC system, it is hard to achieve the optimal operating conditions simply by controlling a single device. Therefore, it is necessary to achieve the global optimal control of the system. The objective function of the optimization problem is to meet the DR signal P_t^{singal} as much as possible for $\forall\, t \in [t_{\mathrm{DR,start}}, t_{\mathrm{DR,end}}]$.

$$\min f(x) = \sum_{\ell\in\mathrm{M}\{\mathrm{ch,fan,pump,cooling\_cycle}\}} P_t^{\ell} - P_t^{\mathrm{singal}}. \tag{17}$$

According to the equipment model in Section 2.1, the chilled water flowrate m_t^{water}, supply air volume m_t^{air}, new air volume $m_t^{\mathrm{air-new}}$, outlet chilled water temperature $T_{\mathrm{o},t}^{\mathrm{ch}}$ and outlet air temperature $T_{\mathrm{o},t}^{\mathrm{coil}}$ can be set as the variables, i.e., $x = [m_t^{\mathrm{water}}, m_t^{\mathrm{air}}, m_t^{\mathrm{air-new}}, T_{\mathrm{o},t}^{\mathrm{ch}}, T_{\mathrm{o},t}^{\mathrm{coil}}]$.

In addition to Eqs. (1)–(9), the constraints consist of the following terms.

(1) Constraint 1: Energy conservation in a coil unit.

$$c_{\text{water}}m_t^{\text{water}}\ |T_{\text{in},t}^{\text{ch}} - T_{o,t}^{\text{ch}}| = c_{\text{air}}m_t^{air}\ |T_{\text{in},t}^{\text{coil}} - T_{o,t}^{\text{coil}}|. \tag{18}$$

(2) Constraint 2: Rated capacity limit of the system.

$$c_{\text{water}}m_t^{\text{water}}\ |T_{\text{in},t}^{\text{ch}} - T_{o,t}^{\text{ch}}| \leqslant Q_{\text{rated}}^{\text{Total}}. \tag{19}$$

(3) Constraint 3: Satisfying the thermal demand of zones Q_t^{total}.

$$c_{\text{air}}m_t^{\text{air}}\ |T_{o,t}^{\text{coil}} - T_{ave}^{room}| = Q_t^{\text{total}}. \tag{20}$$

(4) Constraint 4: Lower and upper bounds for the optimization variables, i.e., the operating state of each commercial air conditioning module.

$$\begin{cases} \underline{m}_t^{\phi} \leqslant m_t^{\phi} \leqslant \overline{m}_t^{\phi},\ \forall\ \phi \in \{\text{water, air, air\_new}\}, \\ \underline{T}_{\text{o},t}^{\iota} \leqslant T_{\text{o},t}^{\iota} \leqslant \overline{T}_{\text{o},t}^{\iota},\ \forall\ \iota \in \{\text{ch, coil}\}. \end{cases} \tag{21}$$

Note that if the response requirement cannot be met by only optimizing the operating state of these modules, the thermal demand can be adjusted with partial comfort sacrifice. Therefore, we can modify the above model by setting Q_t^{total} as a variable to obtain a feasible schedule of the thermal demand.

4.2. Second-Stage Adjustment Method for the Thermal Demand of Zones

4.2.1. Control strategy during the response

According to the thermal demand schedule in Section 4.1, it is necessary to arrange the state of each zone reasonably because the cooling power is different in the cooling duration and warming duration. This approach is similar to the control of a residential air conditioning cluster. In recent years, many efforts have been devoted to studying control strategies for residential air conditioning clusters, and these studies have reference value for the control of the zone states of a commercial CAC system, such as the temperature-priority-list model and dispatching algorithm in [11,38]. Ref. [23] adopts this method to alternatingly and periodically open and close the terminal units of each floor in a commercial building. However, this method only makes sense when the temperature setpoint of each terminal equipment is the same. Thus, it is not suitable for the situation in which customers are more likely to select a temperature setpoint freely in a commercial building. Fortunately, Ref. [35] develops a change-time-priority-list method to solve the above problem. This method is suitable for heterogeneous zones.

The core idea of the change-time-priority-list method is to divide the terminal units into two groups named the cool-down group (Zones I and II in Fig. 6) and the warm-up group (Zones III and IV in Fig. 6). If thermal demand reduction is required, the zones in the cool-down group could be changed to the warm-up group because $Q_{i,\text{cooling}}^{\text{H}}$ is greater than $Q_{i,\text{warming}}^{\text{H}}$, and vice versa.

If we assume that the room temperature variation is controlled within a narrow temperature band $[T_{\min}, T_{\max}]$, the natural change time can be deduced via Eq. (11) as follows:

$$\begin{cases} \tau_i^{\text{change\_cool}} = -\frac{X_i}{B_i}\ln\left(\frac{T_{\max} - (A_i - Q_{i,\text{warming}}^{\text{H}})/B_i}{(A_i - Q_{i,\text{warming}}^{\text{H}})/B_i - T_i^{\text{room}}}\right), \\ \tau_i^{\text{change\_warm}} = -\frac{X_i}{B_i}\ln\left(\frac{T_{\min} - (A_i - Q_{\text{cooling}}^{\text{H}})/B_i}{(A_i - Q_{i,\text{cooling}}^{\text{H}})/B_i - T_i^{\text{room}}}\right). \end{cases} \tag{22}$$

Note that the terminal units in Zones I&III are not suitable for control because these terminal units, whose states have just switched, would be automatically turned back on again soon after manual control. For the terminal units in Zones II&IV, the specific control strategy is shown in Algorithm 1. It is obvious that the wider the temperature band is, the greater the response ability will be. Thus, a new deadband will be set to enhance the ability to respond when a DR event occurs in this strategy.

Algorithm 1. The control strategy based on the change-time-priority list in the response druation

Data: A feasible schedule of the thermal demand Q_t^{follow}
Initialization: Calculate the thermal demand $Q_{t_0}^{\text{total}}$ and the change time before the start of a DR event;
Start Response:
if $t_n \in [t_{DR,start}, t_{DR,end}]$ **then**
  1.0: Calculate the change time of each terminal unit τ_i^{change};
  1.1: Record the units of Zones I/II/III/IV as a series of sets $N_{1/2/3/4}^{t_{n-1}}$. Mark the terminal units of $N_{2/4}^{t_{n-1}}$ where $\tau_i^{\text{change}} < \Delta t$ as $N_{2'/4'}^{t_{n-1}}$, because the states of these units would be switched before the next control;
  1.2: Calculate the thermal demand without control in the next step
$$\begin{cases} Q_{t_n}^{\text{total}} = \sum_{i \in N_1^{t_{n-1}} \cap (N_2^{t_{n-1}} - N_{2'}^{t_{n-1}}) \cap N_{4'}^{t_{n-1}}} Q_{i,\text{cooling}}^{\text{H}} + \\ \sum_{i \in N_{2'}^{t_{n-1}} \cap N_3^{t_{n-1}} \cap (N_4^{t_{n-1}} - N_{4'}^{t_{n-1}})} Q_{i,\text{warming}}^{\text{H}} \end{cases};$$
  if $Q_{t_n}^{total} \le Q_{t_n}^{follow}$ **then**
    2.0: Work out a change-time-priority list of the units in the set $(N_4^{t_{n-1}} - N_{4'}^{t_{n-1}})$ with the order from small to large. To meet the signals as much as possible, select and switch some units to the cool-down group to add thermal demand;
  else
    3.0:Work out a change-time-priority list of the units in the set $(N_2^{t_{n-1}} - N_{2'}^{t_{n-1}})$ with the order from small to large. Reduce the thermal demand by switching some units to the warm-up group from the top of the list;
  end
  4.0: According to the control result in Step 2.0/3.0, calculate indoor temperature of each room at time t_n by equation (11) as well as the final thermal demand of the commercial CAC system.
end

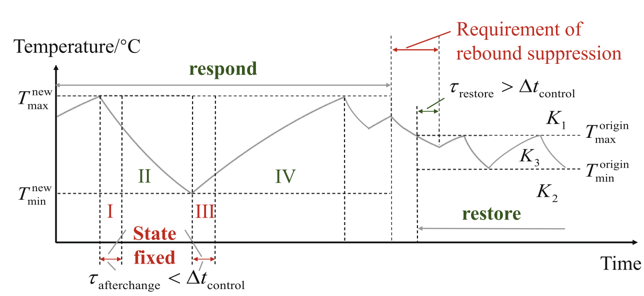

Fig. 6. Division of the control zone according to the change time of indoor unit states: Zones I-II are divided for the DR control while groups K_1-K_3 are for load rebound suppression.

4.2.2. Control strategy after the response

A disorderly setting of the temperature return behavior would result in a larger load rebound after the DR event. Even if the temperature returned to an earlier state, the polymerization thermal demand might fluctuate since the diversity of the load operation state would be destroyed [39,40]. The requirement of rebound suppression is supposed to consider this problem, as shown in Fig. 6. Therefore, a rebound suppression strategy based on the change-time-priority-list method is proposed for the terminal units of the commercial air conditioning system. This strategy is essentially an orderly restoration strategy for the indoor temperature. The goal is to restore the indoor temperature from $[T_{\min}^{\text{new}}, T_{\max}^{\text{new}}]$ to $[T_{\min}^{\text{origin}}, T_{\max}^{\text{origin}}]$ as quickly as possible in case of the load rebound limitation. The control strategy is shown in Algorithm 2.

Algorithm 2. The orderly restoration strategy for load rebound suppression

Data: The requirement of the load rebound suppression $[Q_t^{\text{lower}}, Q_t^{\text{upper}}]$ and the temperature $T_i^{\text{room}}(t_0)$

Control Start:

if $\forall i, \exists T_i^{room}(t_n) \notin [T_{min}^{origin}, T_{\max}^{origin}], t_n \in (t_{DR,end}, \infty)$ **then**

- **1.0**: Divide the terminal units into three groups $K_{1/2/3}^{t_{n-1}}$ according to their indoor temperatures;
- **1.1**: Calculate the change time τ_i^{toupper} and τ_i^{tonormal} for units in $K_1^{t_{n-1}}$, and τ_i^{tolower} and τ_i^{tonormal} for units in $K_2^{t_{n-1}}$; Mark the terminal units of $K_{1/2}^{t_{n-1}}$ where τ_i^{toupper} or $\tau_i^{\text{tolower}} < \Delta t$ as $K_{1'/2'}^{t_{n-1}}$;
- **1.2**: Calculate the rebound thermal demand without control in the next step,

$$\begin{cases} Q_{t_n}^{\text{total}} = \sum\limits_{i \in K_1^{t_{n-1}}} Q_{i,\text{cooling}}^{\text{H}} + \sum\limits_{i \in K_2^{t_{n-1}}} Q_{i,\text{warming}}^{\text{H}} \\ \quad + \sum\limits_{i \in K_3^{t_{n-1}}} Q_{t_n}^{\text{w/o\_control}} \end{cases};$$

- **if** $Q_{t_n}^{lower} \le Q_{t_n}^{total} \le Q_{t_n}^{upper}$ **then**
  - **2.0**: No more operations are needed;
- **else**
  - **if** $Q_{t_n}^{total} < Q_{t_n}^{lower}$ **then**
    - **3.0**: Add the thermal demand by switching some units $M_+^{t_{n-1}}$ in $(K_2^{t_{n-1}} - K_{2'}^{t_{n-1}})$ from the top of the change-time-priority list to meet the limitation,

$$\begin{cases} Q_{t_n}^{\text{lower}} \le Q_{t_n}^{\text{total}} + \sum\limits_{i \in M_+^{t_{n-1}}} \Delta Q_i^{\text{H}} \le Q_{t_n}^{\text{upper}} \\ \text{or } M_+^{t_{n-1}} = K_2^{t_{n-1}} - K_{2'}^{t_{n-1}} \end{cases};$$

  - **else**
    - **if** $Q_{t_n}^{total} > Q_{t_n}^{upper}$ **then**
      - **4.0**: Reduce the demand by switching some units $M_-^{t_{n-1}}$ in $(K_1^{t_{n-1}} - K_{1'}^{t_{n-1}})$ from the change-time-priority list top,

$$\begin{cases} Q_{t_n}^{\text{lower}} \le Q_{t_n}^{\text{total}} + \sum\limits_{i \in M_-^{t_{n-1}}} \Delta Q_i^{\text{H}} \le Q_{t_n}^{\text{upper}} \\ \text{or } M_-^{t_{n-1}} = K_1^{t_{n-1}} - K_{1'}^{t_{n-1}} \end{cases}$$

    - **end**
  - **end**
- **end**
- **5.0**: Calculate indoor temeperature at time t_n as well as the final thermal demand.

end

First, the terminal units are divided into three groups $K_{1/2/3}$: the indoor temperature of K_1 is above $T_{\max}^{\text{origin}}$ and needs to cool down; the indoor temperature of K_2 is below $T_{\min}^{\text{origin}}$ and needs to warm up; and the indoor temperature of K_3 is in the range of $[T_{\min}^{\text{origin}}, T_{\max}^{\text{origin}}]$, which matches the normal operation. The units of $K_{1/2}$ are the control objects.

Then, for K_1, the times to the upper temperature τ_i^{toupper} and the normal temperature τ_i^{tonormal} can be calculated via Eq. (22) for the temperature range $[T_{\text{setpoint}}, T_{\max}^{\text{new}}]$. Similarly, τ_i^{tolower} and τ_i^{tonormal} for K_2 can be obtained for $[T_{\min}^{\text{new}}, T_{\text{setpoint}}]$.

Furthermore, the terminal units of K_1 where $\tau_i^{\text{toupper}} < \Delta t$ are marked as $K_{1'}$, which must be in the cool-down group. It is also well understood that these units would be switched automatically if they were in the warm-up group since their temperature is too close to the upper bound. There is also a set $K_{2'}$ formed in a similar way. The units of $(K_1 - K_{1'})$ are allowed to switch to the warm-up group to reduce the thermal demand, while the units of $(K_2 - K_{2'})$ are allowed to switch to the cool-down group to add the demand.

Therefore, only the units of $(K_1 - K_{1'})$ and $(K_2 - K_{2'})$ can be allowed to control, and two change-time-priority lists can be worked out according to their τ_i^{tonormal} values from small to large. If the thermal demand is larger/smaller than the requirement of rebound suppression, the top units of the lists will be switched preferentially to keep the bottom units in the temperature recovery phase as long as possible.

Combining the proposed reliability evaluation algorithm and control strategy, we then present several numerical results to provide insight into the impacts of some response strategies on the reliability of commercial CAC systems in Section 5.

5. Case studies

The first goal of this section is to examine the effects of the two-stage control strategy, including the demand response and rebound suppression strategies considering customers' thermal demand, failure modes of the load point and the corresponding response requirement. The second goal is to present an illustration of the proposed reliability assessment algorithm and analyze the impacts of different strategies on the reliability performance of the energy supply service.

This section is organized as follows. Section 5.1 introduces the test system and the parameter settings. Section 5.2 illustrates the generation of the response requirement for commercial air conditioning loads. Section 5.3 mainly investigates how DR strategies would improve the system's voltage problem. Section 5.4 studies how customers' comfort and energy supply reliability are affected by different strategies.

5.1. Test system and simulation parameter settings

The case studies are carried out based on the IEEE RBTS-BUS2 system, as shown in Fig. 7. Information such as the customer types, load level, and load point reliability indices can be found in [41]. We modify the original system with load variation profiles that are observed in the real world and assume that air conditioning loads account for 40% to 60% of the peak loads for the commercial buildings, institutions and small users. A different number of CAC systems is configured in these load nodes according to their load level. These air conditioning loads are treated as controllable loads enabling the response.

We adopt commercial CAC equipment data from an architectural model of EnergyPlus, a large office building in Chicago [42], which can also be seen in the appendix A. The rated capacity of this commercial CAC system is approximately 720 kW, and the area of this building is approximately 7200 m^2. We assume that there is an indoor unit every 30 m^2 on average for a total of 240 sets. The mean values of A_i, B_i and X_i are set to 3.45 kW, 43.995 kW/°C, and 157.401 kW/°C, respectively. If all zones require cooling and no reheating is required, we set the cooling power to 0.1 kW/m^2 during the cooling period and 0.05 kW/m^2

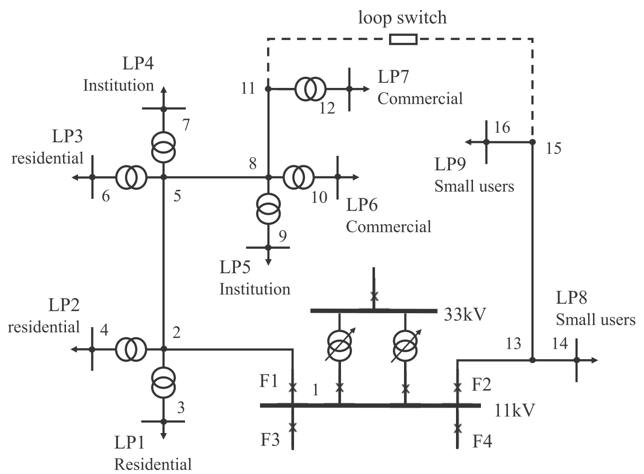

Fig. 7. Initial conditions and network structure of IEEE RBTS-BUS2 system, assuming that air conditioning loads account for 40–60% of the peak loads for the commercial building, institutions and small users.

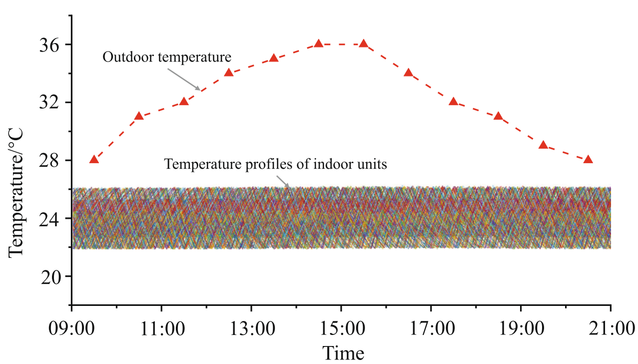

Fig. 8. Outdoor temperature and initial temperature fluctuations of the indoor zones in a commercial building from 9:00 to 21:00.

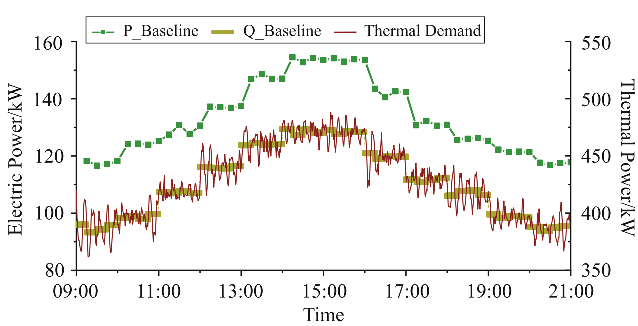

Fig. 9. Baseline construction of the commercial air conditioning load and thermal demand.

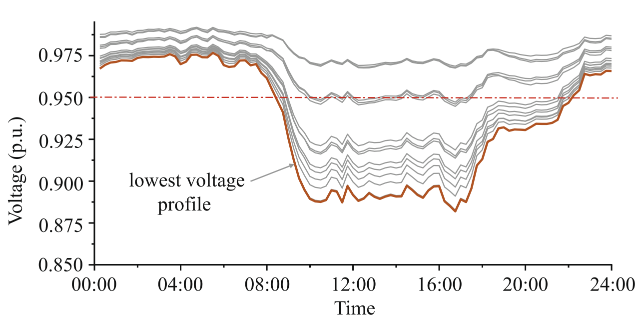

Fig. 10. The voltage distribution of nodes 1–13 when a failure occurs and loads are transferred via a loop switch.

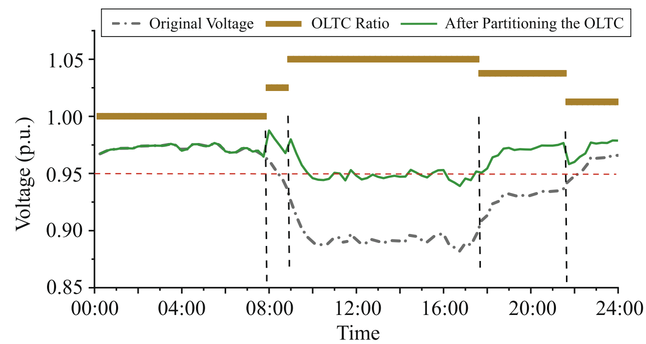

Fig. 11. The OLTC ratio and the voltage after partitioning the OLTC.

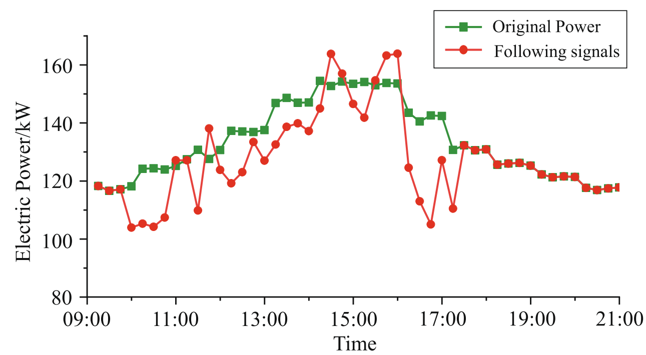

Fig. 12. The original power of commercial air conditioning loads and the following signals of the DR event.

during the warming period. The temperature setpoints of these indoor units are set within [22°C, 26°C] at random, and the deadband is 1 °C. Fig. 8 depicts the initialization process of the indoor temperature.

Furthermore, the electric power baseline and thermal demand baseline can be calculated according to Eqs. (1)–(11), as shown in Fig. 9. It is worth mentioning that the cooling water pump generally adopts a fixed speed pump whose power is constant, and the power of the cooling tower is approximately linear with the cooling water temperature. These two modules generally account for less than 5% of the total power. Therefore, their total power is considered to be a fixed value.

In addition, the on-load tap changing transformer (OLTC) is adopted with ±4 grades, and each grade is adjusted by 0.0125 p.u. in this case study system. The loads of Feeder 1 and Feeder 2 can be flexibly transferred through the loop switch if a failure occurs on the feeder line. However, long lines and heavy loads after transferring might result in a low voltage problem, as shown in Fig. 10. At this point, some controllable resources could be utilized to relieve the voltage violation, and a DR event to commercial air conditioning loads might occur.

5.2. Determining the following signal of a DR event for commercial air conditioning loads

Usually, the OLTC has a decisive effect on voltage regulation, and its setting is determined first. A tap position (modeled as an increment of the secondary voltage setpoint) should be able to keep the voltages on all feeders within the voltage limits in a time segment. However, the equipment to be adjusted should be discrete elements due to the daily manipulation limits stemming from lifespan considerations. The optimal partitioning technique can be used to divide a series of data into a designated number of segments based on maximum homogeneity. The similarity of system voltage profiles within a segment of time suggests little need to change the OLTC's settings. Interested readers can refer to [43] for details.

After traversing all the fault transfer scenarios, the voltage problem

Table 1
Simulated events at 12:00 pm.

Simulation	Voltage	DR signals	Equipment operation status	Power demand	Thermal demand
W/O control	0.943 p.u.	119.179 kW	{0.031 m^3/s, 26.67 m^3/s, 6.97 m^3/s, 6.31 °C, 11.39 °C}	137.259 kW	440.414 kW
W/O comfort sacrifice	0.945 p.u.	119.179 kW	{0.029 m^3/s, 24.04 m^3/s, 3.48 m^3/s, 5.59 °C, 10.00 °C}	130.214 kW	440.414 kW
Proposed strategy	0.950 p.u.	119.179 kW	{0.025 m^3/s, 21.94 m^3/s, 3.48 m^3/s, 6.06 °C, 10.01 °C}	119.284 kW	402.015 kW

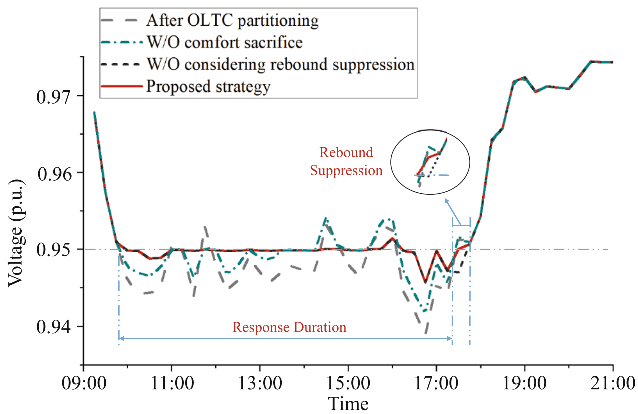

Fig. 13. The voltage profiles of the distribution system under different response strategies, including the control after OLTC partitioning, W/O comfort sacrifice, W/O rebound suppression and with the proposed strategy.

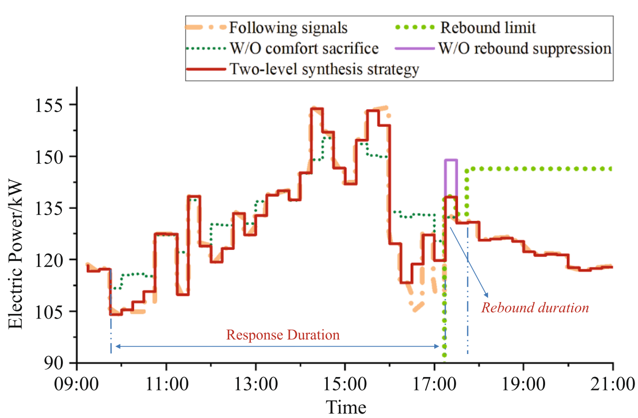

Fig. 14. The power profiles of the commercial air conditioning system.

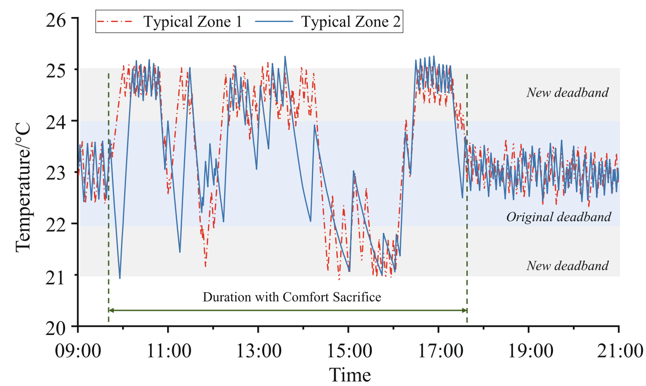

Fig. 15. The temperature of typical zones.

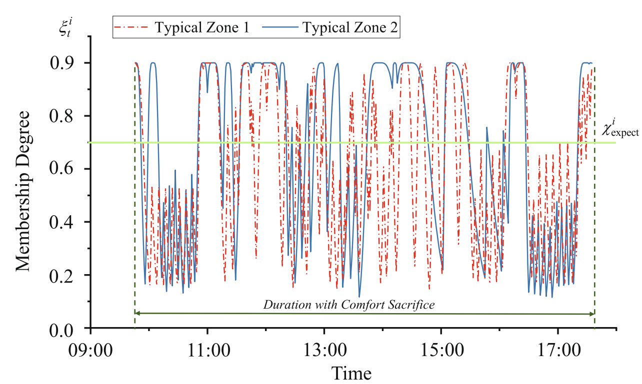

Fig. 16. The comfort membership degree of typical zones.

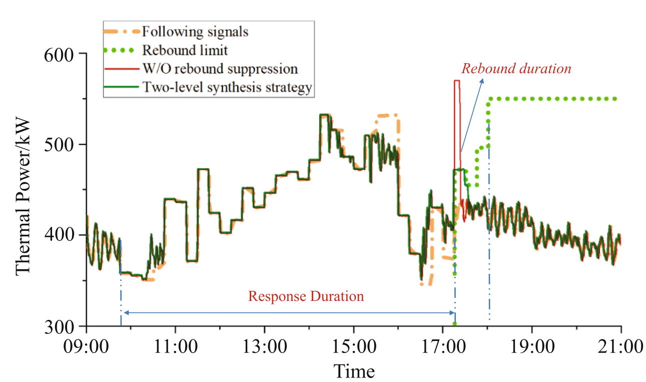

Fig. 17. Impact of different strategies on the thermal demand.

Table 2
Low voltage improvement under different DR strategies.

Strategies	Qualified rate	Lowest voltage	Time to suffer from low voltage
W/O control	42.71%	0.882 p.u.	4.48 h
After OLTC partitioning	77.08%	0.939 p.u.	1.79 h
W/O comfort sacrifice	84.38%	0.941 p.u.	1.22 h
W/O rebound suppression	89.58%	0.943 p.u.	0.81 h
Proposed strategy	93.75%	0.943 p.u.	0.49 h

cannot be solved by only adjusting the OLTC tap position when a failure occurs in feeder 13 if the qualified voltage is set to 0.95, as shown in Fig. 11. The voltages at 9:45–17:15 need to be further improved. The reliability indices can be found in [41]: the fault rate λ is 0.161 f/yr, and the outage time r is 20.58 h. Thus, the probability $\psi(t_{\mathrm{DR,start}}, t_{\mathrm{DR,end}})$ that a DR event occurs can be simply regarded as the fault rate $\lambda/365$. Finally, we can utilize the sensitivity between the power flow and node voltage to determine the power following signals of commercial air conditioning loads in each node. The objective is to regulate the load adjustment of each node as evenly as possible. Due to the page limitation, the details of this solving method are not presented here. The final following signal for each air conditioning system is shown in Fig. 12.

5.3. DR effect on the system voltage problem

First, how the proposed control strategy for commercial air conditioning loads in Section 4 is applied will be demonstrated. For

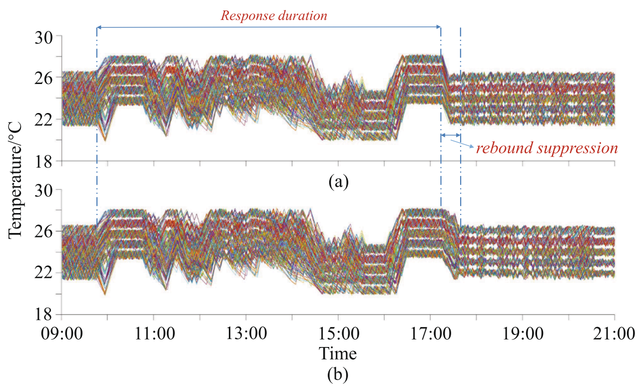

Fig. 18. Impact of different strategies on room temperature profiles: (a) W/O rebound suppression; (b) Two-level synthesis strategy.

Table 3
Reliability performance under two control strategies.

Types	Indices	W/O rebound suppression	Two-level synthesis strategy
Institution	ATDUI	0.39147‰	0.36420‰
	QENS	12.94 kWh	13.10 kWh
Commercial	ATDUI	0.19482‰	0.21503‰
	QENS	2.90 kWh	3.01 kWh
Small user	ATDUI	0.13600‰	0.15438‰
	QENS	4.63 kWh	4.92 kWh

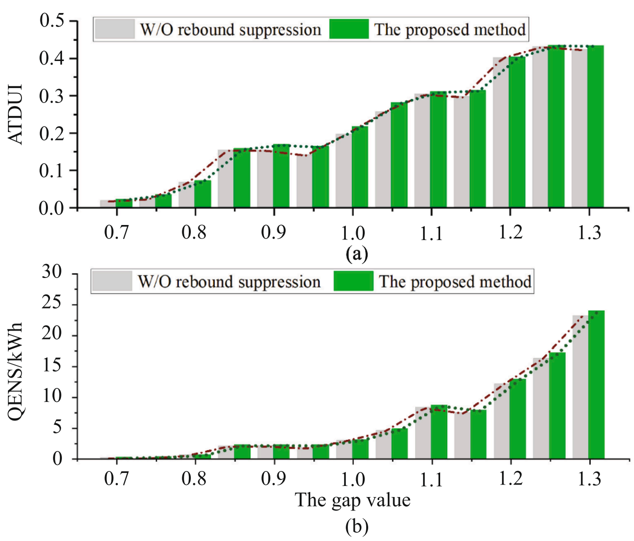

Fig. 19. Results of sensitivity analysis: (a) ATDUI performance with varying following signals; (b) QENS performance with varying following signals.

example, the system suffers voltage violation risks at 12:00 pm. The results shown in Table 1 demonstrate how the undesired voltage is relieved. The fourth line, named equipment operation status, shows the operation status of a commercial air conditioning system, i.e., variables $x = [m_t^{\mathrm{water}}(\mathrm{m^3/s}),\ m_t^{\mathrm{air}}(\mathrm{m^3/s}),\ m_t^{\mathrm{air-new}}(\mathrm{m^3/s}),\ T_{o,t}^{\mathrm{ch}}(°\mathrm{C}),\ T_{o,t}^{\mathrm{coil}}(°\mathrm{C})]$.

The first stage of control is to optimize the operating state of the internal equipment in a commercial CAC system collaboratively with no thermal demand adjustment. By contrast, the proposed method, i.e., a two-level synthesis strategy, can sacrifice some customers' comfort to meet the qualified rate. Note that to enhance the adjustment ability of the thermal demand, a wider temperature band of 4 °C is adopted in working out the change-time-priority lists in this paper.

By repeating the strategies for all time snapshots, we will be able to obtain an estimation of the operation performance of the system. The voltage profiles of the test system and the electric power of the commercial air conditioning system under different control strategies are shown in Fig. 13 - Fig. 14. Two issues are stressed in the figures.

(1) The voltage violation of the distribution can be improved to different degrees via the participation of air conditioning loads in the DR, as shown in Fig. 13. The voltage qualified rates under the four scenarios are shown in Table 2. In addition, by comparing the solid red line and dotted line with the short-dot chain line in Fig. 14, it is obvious that the power demand under the two-level synthesis strategy can be more responsive to the following signals than that of the strategy without comfort sacrifice. As a result, the voltage can be further improved at some times, and the adjustable range of the air conditioning loads can be greatly expanded with partial comfort sacrifice.
(2) A significant spike in power consumption will be triggered when no limitation is imposed on the air conditioning system (the solid purple curve in Fig. 14). Although the thermal storage capacity of the air conditioning system can be fully exploited by adjusting the thermal demand, a load rebound might occur by restoring the room temperature and cause another voltage violation risk (the magnified part in Fig. 13). The DR event needs to last until 17:45 due to the need to further suppress the load rebound, which was supposed to end at 17:35. However, the time to suffer from low voltage is the smallest considering the rebound suppression, as shown in Table 2. This observation explains why we make an orderly restoration strategy. The results of the power demand can also prove that the proposed method can better suppress the load rebound within the acceptable range of the power system.

5.4. Effect of the control strategies on reliability

First, this section demonstrates the key procedures to identify whether the LoL event proposed in Section 3.2 occurs step-by-step using some typical DR results.

(1) Determine the DR start/end time, e.g., the DR event starts at 9:45, and according to its failure mode, the end time is assumed as 17:45, as shown in Fig. 14.
(2) Obtain the room temperature profiles of these zones, as shown in Fig. 15. The initial set temperature is 25 °C, and the initial deadband is 2 °C. Once the demand response starts, a new deadband 4 °C is adopted. The moment at the gray band is the time that does not meet the customers' original demand.
(3) Calculate the membership degree between temperature and comfort by Eq. (12). Assuming that the values of α_t, β_t, χ^i_{expect} are 1.5, 2.5, 0.7, the profiles of membership degree can be seen in Fig. 16.
(4) Determine if a "loss of load" event occurs. The percentage standard of discomfort time $\delta = 50\%$. (1) For Zone 1, according to (13) we have $\varepsilon = 1$ since $286 - 50\% \times 470 > 0$, which means a LoL event occurs; (2) For Zone 2, we have $\varepsilon = 0$ since $223 - 50\% \times 470 < 0$, which means a successful energy supply service is provided to the customer in the DR duration.

Further, according to the above analysis, the reliability performance of commercial air conditioning loads will be affected when the thermal demand is adjusted. This effect means that the power supply reliability will not decrease by the operating state optimization of the internal equipment without comfort sacrifice. Thus, we demonstrate the thermal demand and the room temperature of two strategies with partial comfort sacrifice to calculate the reliability.

As shown in Fig. 17 - Fig. 18, the response requirement can be met most of the time during the response, and the indoor temperature can be limited within a prescribed room temperature deadband. However, without a reasonable rebound suppression strategy, a significant spike in the thermal demand will far surpass the limit, and this finding is deduced according to the above electric power rebound limit (the green dotted line in Fig. 17). This phenomenon can be explained according to the room temperature profiles in Fig. 18. Once the DR event is over, most of the indoor temperature is higher than the original setpoint due to the change in the temperature deadband. Hence, all terminal units will be fixed in the cool-down group, which will greatly increase the cooling demand. However, under the control of the proposed strategy, the temperature can be restored in an orderly manner according to the restrictions.

The reliability performance under these two control strategies is shown in Table 3. In this case, we set the comfort membership limit $\chi^{i}_{\text{expect}} = 0.7$ and the percentage standard of discomfort time $\delta = 50\%$. The parameters of the fuzzy membership degree vary with the load types, assuming that institutions prefer a comfortable temperature, commercial loads are second, and small industrial loads are the worst. The average values of $\{\alpha_t, \beta_t\}$ are $\{1.4, 2.2\}$, $\{1.5, 2.5\}$ and $\{1.55, 2.3\}$, respectively. From the results of the reliability indices, although the load rebound can be effectively suppressed, it will prolong the time needed for the room temperature to be fully restored and might increase customers' discomfort.

Beyond that, lowering/raising the response requirement, i.e., reducing/magnifying the gap between the following signal and the power baseline, requires an air conditioning system with different adjustable ranges. Changing the gap between the following signal and the power baseline may be considered a practical way by DSOs to realize targeted reliability performance. Therefore, the sensitivity analysis in this section shows how energy supply service reliability would benefit from this approach.

The air conditioning system of commercial loads is taken as an example. Setting the gap value ρ between the following signals and the baseline load shown in Fig. 12 as 100 %, we change the gap value ρ from 70 % to 130 %. The ATDUI and QENS indices of the commercial loads are given in Fig. 19(a) and Fig. 19(b), respectively. According to the results, varying the following signals can result in different reliability performance. This approach can help guide DSOs to issue reasonable DR commands according to the reliability requirements of different types of loads for energy supply services. Additionally, this can be used for the commodity function and value of air conditioning services to work out some incentive methods for which we have conducted a preliminary study using value engineering theory in [44].

6. Conclusion

By considering commercial air conditioning loads as controllable and adjustable, this paper proposes a systematic reliability evaluation algorithm of the energy supply processes for distribution systems. A two-stage demand response strategy is developed considering a necessary condition for reliability evaluation. Case studies are conducted to verify the effectiveness of our proposed method. The major conclusions could be derived as follows:

(1) A clear definition of "Loss of Load" for commercial air conditioning loads can reflect the impact of a demand response strategy on customer satisfaction. The acceptable standard can be formulated according to the customers' diversified preferences in our method.
(2) Through combining the global optimization of the equipment and the thermal power consumption, the adjustable potential of air conditioning loads can be fully utilized. Meanwhile, the control methods for both the demand response and rebound suppression are considered. In the case of a low voltage network, the results demonstrate the positive effect of our control strategy as well as the impacts of rebound suppression on reliability. Although the energy supply reliability would partly decline, the system would not suffer a second violation risk caused by the load rebound.
(3) The customer-oriented and energy-oriented indices are developed to quantify load reliability. Sensitivity analysis of these reliability indices under different response depths would act as important insights to guide distribution system planning and operation.

In future works, we will focus on establishing an aggregation strategy and developing a comprehensive indices system for the dispatch of large-scale commercial air conditioning loads.

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Appendix A. Appendix

The equipment parameters of a commercial CAC system can be found in the Table A1.

Table A1
Equipment parameters of a commercial CAC system.

Chilled water rated outlet temperature	Chilled water design flowrate	Rated capacity
5.56 °C	0.031 m^3/s	721.00563 kW
Rated COP	Min load rate	Max load rate
6.04	0.19	1.04
Min chilled water outlet temperature	Max chilled water outlet temperature	Pump rated water flowrate
2 °C	10 °C	0.31 m^3/s
Pump rated power	Fan design pressure	Fan efficiency
7.92303 kW	1 kPa	0.7
Max fan air volume	Min fan air volume	Coefficient a_1
40.64 m^3/s	6.37 m^3/s	0.4418712
Coefficient a_2	Coefficient a_3	Coefficient a_4
0.0492	−0.002190707	0.3899
Coefficient b_1	Coefficient b_2	Coefficient b_4
0.5059102	−0.0289	0.0003346388
Coefficient c_1	Coefficient a_3	Coefficient c_2
0.6107	0.1879418	0.3562862
Coefficient c_3	Coefficient d_1	Coefficient d_2
0.4540392	0.35071223	0.30850535
Coefficient d_3	Coefficient d_4	Coefficient e_1
−0.54137364	0.87198823	1

Appendix B. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.apenergy.2019.113547.

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